

Aegis – Product Backlog – Group 18

Charles Law, Jayden Deason, Bilsha Das, Michael Lenkeit

Problem Statement

Current task management tools like are overloaded with features that are overwhelming and have a steep learning curve, especially for small teams and individual users who need a straightforward way to manage tasks. The Aegis project management system will focus on simplicity, prioritizing core features like task creation, assignment, and progress tracking without unnecessary complexity. Its user interface will be simple and intuitive, making it accessible for those unfamiliar with using similar systems.

Background Information

- **Background Information:** An essential part of any teamwork-based environment is project management, including planning, executing, and finalizing projects while allocating teams and resources.
- **Similar Sites:**
 - Sites such as Microsoft Teams, Slack and Discord are used to fill a similar purpose, giving an organization the ability to self-organize into smaller groups that each focus on specific tasks, but are designed for a much broader range of use cases and lack the focus that Aegis will have on managing multiple tasks efficiently.
 - Sites such as Asana, Monday.com, Miro and more are used for project collaboration and repository purposes but could benefit from the team management and communication aspects that Slack and Discord offer.
 - Aegis aims to combine both the communication and project management aspects of these apps in an integrated and intuitive way that makes task creating, tracking and sharing simple.
- **Limitations:** These existing applications offer similar project management features but can be overwhelming for new users with their steep learning curve and a stupendous number of features. Other limitations include a high subscription fee. Our app will solve this issue by offering a simple, yet intuitive interface to enhance the user's experience while maintaining an affordable cost.

Requirements

Functional:

All users:

1. As an account holder, I would like to create an account for Aegis.
2. As an account holder, I would like to log into my Aegis account from different devices.
3. As an account holder, I would like my login information to be stored in a secure fashion.
4. As an account holder, I would like to be able to reset my password with an email.
5. As an account holder, I would like to be able to opt into using two-factor authentication and receive a code in an email to log into my account.
6. As an account holder, I would like to sort my tasks differently based on urgency, assigner, and project in the forms of a to-do list, kanban board or calendar.
7. As an account holder, I want to switch between different views (to-do list, kanban board, calendar) to organize my tasks according to my preference
8. As an account holder, I want to filter tasks by urgency levels (e.g., high, medium, low) to prioritize my work.
9. As an account holder, I want to sort tasks by the person who assigned them to me, so I can manage expectations and communication.
10. As an account holder, I want to group tasks by project to see all related tasks in one place.
11. As an account holder, I want to search my list of tasks for those containing specific phrases or that have specific urgency levels.
12. As an account holder, I would like to view project progress reports so that I can assess team performance.
13. As an account holder, I would like to be able to message other account holders, through direct messages or in larger group chats (specific to a task or a project).
14. As an account holder, I want to initiate direct messages with other users to discuss specific issues privately.
15. As an account holder, I want to create group chats for specific tasks to facilitate team collaboration.
16. As an account holder, I want to create project-specific chat rooms to discuss broader project goals and updates.
17. As an account holder, I want to receive notifications for new messages to stay updated on communications.
18. As an account holder, I want to exist in multiple organizations, projects and access multiple project-specific chats within each server.
19. As an account holder, I want the ability to delete any messages that I have sent.
20. As an account holder, I would like my documents, files, and chats to be stored securely and protected from SQL injection attacks.

Task Assignee:

21. As a task assignee, I would like to view a list of all the tasks I have been assigned.
22. As a task assignee, I want to view a calendar that shows all the tasks listed on their respective due dates.
23. As a task assignee, I would like to update the completion status of my individual tasks
24. As a task assignee, I would like the ability to notify my task assigner when I make updates to my tasks.
25. As a task assignee, I would like to attach documents to specific tasks that would be shared with the account holder who assigned me the task.
26. As a task assignee, I want to upload documents directly to a task to provide necessary resources or updates.
27. As a task assignee, I want to view and download documents attached to tasks to access shared resources.
28. As a task assignee, I want to receive notifications when new documents are attached to tasks I am assigned to.
29. As a task assignee, I would like to receive email notifications before projects are due.
30. As a task assignee, I would like to be able to select the amount of time before projects are due that I get a reminder notification (i.e. I get an email 3 days beforehand).
31. As a task assignee, I would like to send multiple invitations to users join my project

Task Assigner:

32. As a task assigner, I want to assign different tasks to one or multiple users that allow different levels of permissions to view the task, modify the task or assign the task.
33. As a task assigner, I would like to add or remove people from the task.
34. As a task assigner, I would like to change the permission level (view, modify or assign) of people assigned to the task.
35. As a task assigner, I would like to view a list of all the tasks I have assigned others, with the option to filter by the urgency or status of each project
36. As a task assigner, I would like to assign various levels of importance to tasks.
37. As a task assigner, I want to edit tasks, including the title, text, repository, importance level or deadline.
38. As a task assigner, I want to delete tasks and notify the people assigned to them.
39. As a task assigner, I would like the system to automatically create a group chat that includes everyone assigned to the task if one does not already exist, and to update the permissions of each member as people are added or deleted.
40. As a task assigner, I should be the only one who can add users to my task.
41. As a task assigner, I should be able to make someone else the task assigner if necessary.

Project Manager:

42. As a project manager, I would like to create projects so I can organize my team efficiently.
43. As a project manager, I would like to edit projects so I can organize my team efficiently.
44. As a project manager, I would like to delete projects so I can organize my team efficiently.
45. As a project manager, I want to define project goals and objectives during creation to guide the team.
46. As a project manager, I want to update project details, such as deadlines and descriptions, to reflect in scope.
47. As a project manager, I want to archive completed projects to maintain a clean workspace while preserving historical data.
48. As a project manager, I would like to be able to assign users to a project, with clear communication with the users.
49. As a project manager, I would like the ability to delete any tasks within the project.

Admin:

50. As an admin, I would like to view a list of all users in my organization.
51. As an admin, I would like to manage users' roles so I can control their access to certain features.
52. As an admin, I would like to assign roles such as user, task assigner, project manager, and admin to control feature access
53. As an admin, I would like to be able to add or remove users to my organization.
54. As an admin, I would like to access all private messages between people in my organization, including all deleted messages.
55. As an admin, I want to update user roles as needed to adapt to organizational changes
56. As an admin, I want to notify users of roll changes to ensure they are aware of their new responsibilities

Non-Functional:

Performance

- The system should respond to user actions within one second to ensure a smooth user experience.
- The app should support at least one-hundred simultaneous requests to handle peak usage times.
- The application dashboard should load within five seconds so users can quickly access their projects.

- Task updates should be reflected in less than one second so users can immediately see requests

Scalability

- The system should be able to scale horizontally to accommodate at least 500 user accounts.
- As the system grows, it should not suffer from performance degradation

Usability

- The app should have an intuitive interface that new users can learn within 30 minutes.
- Navbar: Allows seamless search between all tasks, projects, document repositories, person and chats that you have access to in one search bar.
- Task Tracker Views:
 - Calendar: Visually sorts all tasks based on their due dates. Provides support for weekly, monthly and yearly views, with the ability to look at past and future time periods; based on chronological urgency level
 - To-do list: Lists all tasks with a brief description that allows for users to sort by various attributes, including due date, importance level, task assigner, and date assigned.
 - Kanban Board: Tasks in the form of tiles that move across various columns filtered based on importance level

Security

- User data should be encrypted in transit and at rest to protect privacy.
- The app should implement two-factor authentication for enhanced security.
- User's session should automatically log out after fifteen minutes of inactivity.
- All user actions should be logged to audit activities.
- Data should be filtered before adding to the DB to prevent injection attacks.
- Prevent malicious SQL input from being included in executed queries